



HIGH PERFORMANCE COMPUTING 101

Data Transfer & Best Practices (Module 4)

Module 4 — Lesson 1

Data Transfer



Overview

- Choosing a transfer method
- SCP & SFTP
- Globus
- SDA with HSI and HTAR
- Archiving files to the SDA



Choosing a transfer method

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Decision table

Scenario	Tool
Large files (over 1 GB) between IU systems	Globus
Personal computer to or from an IU system, any size	Globus
Google Drive or OneDrive to or from an IU system	Globus
Quick one-shot copy between two Linux hosts	SCP
Interactive file browsing on a remote system	SFTP
Interactive shell access to the SDA	HSI
Bundling many small files into SDA archives	HTAR



Constraints

Two constraints override everything in the table:

- Protected health information (PHI), any direction: Globus or SFTP only
 - HSI and HTAR do not encrypt in transit and must never be used for PHI
- Off-campus to the SDA without an exemption: Globus or SFTP
 - HSI and HTAR are on-campus tools; off-campus requires an exemption from store-admin@iu.edu



SCP & SFTP

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SCP — one-shot copy

- Copies a file in a single command, no interactive session
- Best for quick, small copies when you know the exact paths
- Remote paths take the form ``username@hostname:path``
- Upload to Quartz:
 - `$ scp mydata.tar.gz username@quartz.uits.iu.edu:/N/slate/username/`
- Download from Quartz:
 - `$ scp username@quartz.uits.iu.edu:/N/slate/username/results.csv .`
- ``-r`` copies a directory recursively
- Two-Step Login (Duo) required; SSH public key authentication available via the SSH public key authentication agreement
- No resume: a failed transfer restarts from the beginning



SFTP — interactive browse

- Interactive session: browse the remote tree, then transfer
- Best for small transfers when you want to navigate before copying
- Start a session:
 - **\$ sftp username@quartz.uits.iu.edu**
- In-session commands: ``ls``, ``cd``, ``pwd``, ``put localfile``, ``get remotefile``, ``rename old new``, ``exit``
- ``lls``, ``lcd``, ``lpwd`` act on your local side
- Connect to the SDA: **\$ sftp username@sftp.sdarchive.iu.edu**
- GUI clients exist: WinSCP (Windows), Cyberduck (macOS)
- Two-Step Login (Duo) required; SFTP resumes interrupted transfers



Globus

- Choosing a transfer method
- SCP & SFTP
- **Globus**
- SDA with HSI and HTAR
- Archiving files to the SDA



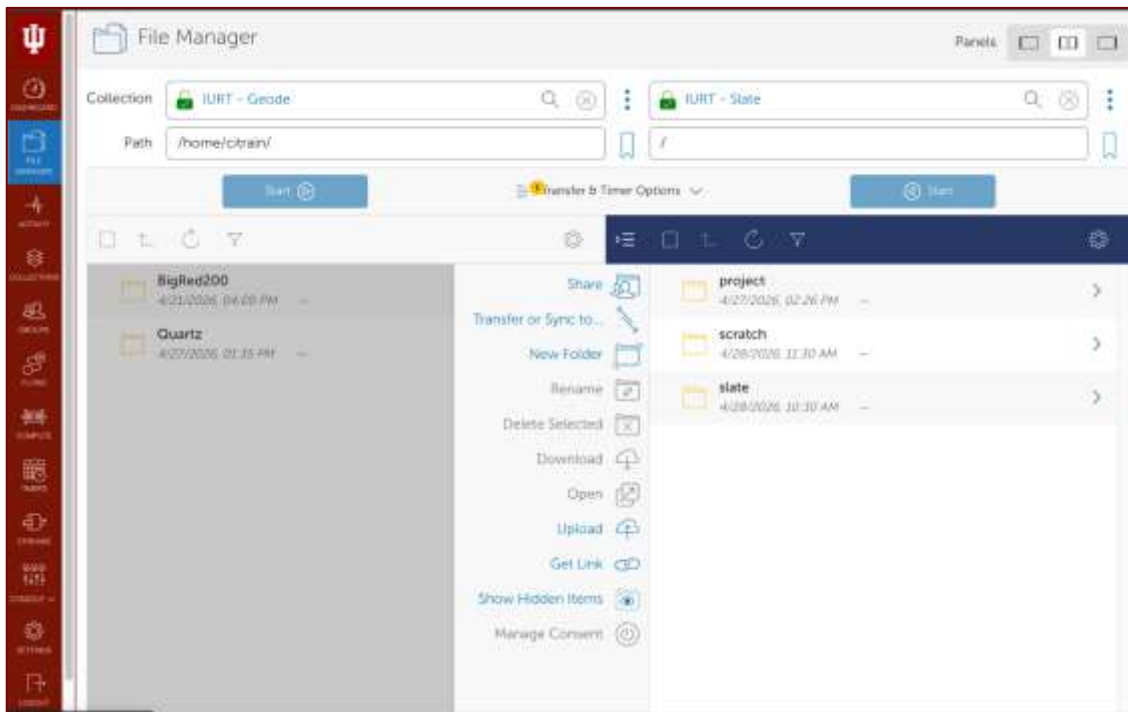
Globus and Moving data

- Globus provides a mechanism similar in usage to ftp/sftp
 - different in that there is no "local vs remote" distinction
 - Globus web client allows you to schedule transfer between "endpoints"
 - Globus endpoints for IU HPC are named "IURT - *something*"
 - e.g. "IURT - Scholarly Data Archive", "IURT - Slate"
 - IURT - OneDrive allows access to your OneDrive account
 - Desktop client you can install to make your local computer an endpoint



Globus

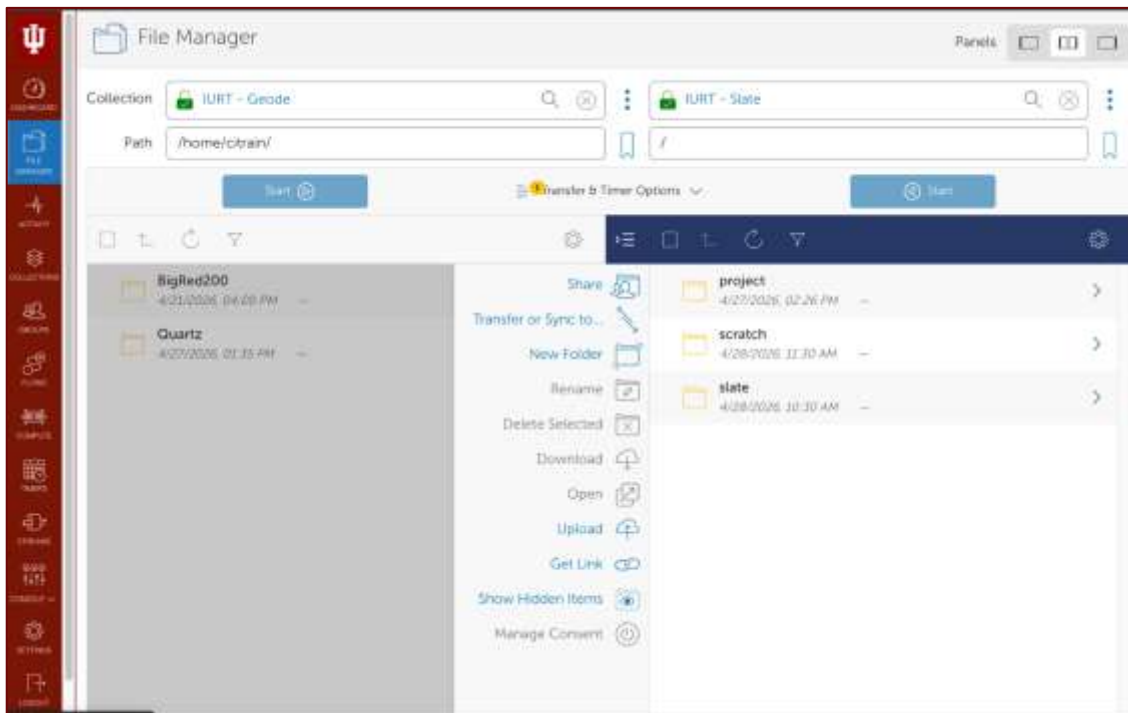
- Open the URL <https://globus.iu.edu/>
- Log in with your IU credentials & Duo
- Default window is File Manager
 - Panels allows split view
- Search IURT for collections





Globus

- "IURT - Geode" maps to /N/u/username by default
 - your home dirs
- "IURT - Slate" maps a directory view above
 - /N/project
 - /N/scratch
 - /N/slate
- Point and click through folders to your data
 - select and transfer





Globus

- Typically, moving data from Geode to Slate would be much faster using the file system on an HPC login node
 - These two endpoints were shown for illustrative purposes
- "IURT - Scholarly Data Archive" is the Globus endpoint for your SDA account
 - Globus is an excellent tool to transfer data from Geode or Slate to SDA
 - Note: bundle small files (< 1 MB) into larger tar or zip files first on the filesystem
 - You can store huge file sizes on SDA, but know tar & zip limits



Globus

- Archive your data and work to SDA frequently enough to protect your progress
 - definitely at important milestones
- Use a file-naming strategy and your notes to keep track of what is in any given tar/zip file you move to SDA.
 - Include a date, so that you can keep updated backups
 - Delete things that are redundant or which you don't need
- Clean up file system space when you won't be using it for a while



SDA with HSI and HTAR

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HSI — interactive shell to the SDA

- HSI opens an interactive shell on the SDA
- Available on all IU research supercomputers; load the module first:
 - **\$ module load hpss**
- Start a session with ``hsi``, then use in-session commands:
 - ``ls``, ``cd``, ``pwd``, ``mkdir``, ``put localfile``, ``get remotefile``, ``mv old new``, ``exit``
- From Research Desktop (RED):
 - Applications > Storage > Scholarly Data Archive



HSI — interactive shell to the SDA

- On-campus only; off-campus requires an exemption from store-admin@iu.edu
- Not PHI-safe: HSI does not encrypt in transit, use Globus or SFTP for PHI
- SDA deletions are permanent and irreversible



HTAR — bundle many files to tape

- HTAR bundles many local files into a single ``.tar`` archive directly on the SDA
 - no intermediate local archive needed
- The right tool when you have many small files, which the SDA handles poorly one by one
- Load the module first: `$ module load hpss`
- Create an archive from the current directory:
 - `$ htar -c -f my_archive.tar *`
- List contents: `$ htar -tf my_archive.tar`; extract: `$ htar -x -f my_archive.tar`



HTAR — bundle many files to tape

- Limits (more restrictive than the SDA itself):
 - max individual file size 68 GB; up to 1 million files per archive
 - max directory path 154 characters; max file name 99 characters
- On-campus only; not PHI-safe (use Globus or SFTP for off-campus or PHI)



SDA prep rules

- Tape is unforgiving of many tiny files: bundle files under 1 MB into ``tar`` or ``zip`` first
- Use ASCII-only characters in paths and file names (0x20 to 0x7e); rename anything with Unicode first
- Archive at milestones: before submitting a paper, after a major analysis, at phase changes



SDA prep rules

- Date your archive file names so you can keep updated versions and find the latest
- Keep a separate contents log: the SDA can't preview a ``.tar`` without retrieving it
- Delete redundant or superseded data from active file systems to stay within quota



Archiving files to the SDA

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Archiving files to the SDA

1. On Quartz, create a multi-gigabyte test file in your Slate directory:

```
$ cd /N/slate/username
```

```
$ dd if=/dev/urandom of=testfile.dat bs=1M count=2048
```

1. Record a checksum so you can compare later:

```
$ sha256sum testfile.dat
```

1. Send the file to the SDA over SFTP:

```
$ sftp username@sftp.sdarchive.iu.edu
```

in the session: ``put testfile.dat``

1. Rename the file on the SDA, still in the SFTP session, then ``exit``:
`rename testfile.dat testfile_2026-06-12.dat`



Archiving files to the SDA

5. Pull the renamed file back with Globus (<https://globus.iu.edu/>):
source `IURT - Scholarly Data Archive`, destination `IURT - Slate`
at `/N/slate/username/`
select `testfile\_2026-06-12.dat`, then Start
6. Confirm the round trip matched by checksum:
\$ sha256sum testfile_2026-06-12.dat
compare against the value from step 2
7. Clean up the temporary files:
on Quartz: **\$ rm testfile.dat testfile_2026-06-12.dat**
in HSI or Globus: delete `testfile\_2026-06-12.dat` from the SDA



Reflection

- You can now:
 - Choose a transfer tool by matching size, source, and destination
 - Copy between hosts with SCP and browse and transfer with SFTP
 - Move large, recurring, and cloud transfers with Globus
 - Archive to and retrieve from the SDA with HSI and HTAR
- Next lesson: Module 4, Lesson 2 — Managing Resources & Best Practices
- Be sure to complete the reflection for this lesson